

AP CALCULUS AB - UNIT 1

Guided Notes 1.5: Determining Limits
Using Algebraic Properties of Limits

Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we apply constant, sum, difference, product, quotient, power, root, and composition laws?

- Apply constant, sum, difference, product, quotient, power, root, and composition laws.
- Check every hypothesis before using a limit law.
- Separate known limit values from function values at the target.
- Justify direct substitution through continuity and domain conditions.

Key Knowledge, Vocabulary, and Notation

Linearity

$$\lim (af + bg) = a \lim f + b \lim g \text{ when the component limits exist.}$$

Products and powers

$$\lim (fg) = (\lim f)(\lim g) \text{ and } \lim [f(x)]^n = (\lim f)^n.$$

Quotient condition

$$\lim \frac{f}{g} = \frac{\lim f}{\lim g} \text{ only when the denominator limit is nonzero.}$$

Even roots

For real-valued limits, the landing value must lie in the appropriate domain; endpoint cases may be one-sided.

Composition

If $g(x) \rightarrow L$ and the outer function is continuous at L , then $F(g(x)) \rightarrow F(L)$.

Concept Development and Strategy

1. Write the limit of each changing component before combining them.
2. A failed hypothesis means a law is inconclusive, not that the limit automatically fails.
3. Direct substitution is a theorem-based shortcut for continuous functions at allowed inputs.

Shared Representation / Data

	$\lim_{x \rightarrow 2} f(x)$	$\lim_{x \rightarrow 2} g(x)$	$\lim_{x \rightarrow 2} h(x)$
Value	4	-3	0

Worked Example: Linear combination

Given $\lim_{x \rightarrow a} f(x) = 5$ and $\lim_{x \rightarrow a} g(x) = -2$, find $\lim_{x \rightarrow a} [3f(x) - 4g(x) + 1]$.

Answer and Reasoning

$$3(5) - 4(-2) + 1 = 24.$$

Worked Example: Nested structure

If $f(x) \rightarrow 3$ and $g(x) \rightarrow -1$, evaluate $\lim \sqrt{2f(x)^2 - g(x)}$.

Answer and Reasoning

The inside approaches $2(3)^2 - (-1) = 19 > 0$. The square-root function is continuous there, so the limit is $\sqrt{19}$.

Worked Example: Failed quotient hypothesis

If $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$, what does the quotient law say about $\lim f(x)/g(x)$?

Answer and Reasoning

Nothing conclusive. The denominator-limit hypothesis fails. The actual quotient limit might be finite, infinite, or nonexistent depending on the functions.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Using the shared limit data, find $\lim_{x \rightarrow 2} [f(x) + g(x)]$.

2. Using the shared limit data, find $\lim_{x \rightarrow 2} [2f(x) - 5g(x)]$.

3. Using the shared data, evaluate $\lim_{x \rightarrow 2} \sqrt{f(x) + 5}$.

4. Using the shared data, evaluate $\lim_{x \rightarrow 2} \frac{f(x)-1}{g(x)+5}$.

5. If $f(x) \rightarrow 4$ and $g(x) \rightarrow 0$, can the quotient law evaluate $\lim f(x)/g(x)$? Explain precisely.

6. Assume $f(x) \rightarrow -4$. Determine whether $\lim \sqrt{f(x) + 4}$ is automatically valid as a real two-sided limit.

7. Let $f(x) \rightarrow a$ and $g(x) \rightarrow b$. Find a condition on a, b that guarantees $\lim \frac{f(x)^2 - g(x)^2}{f(x) - g(x)} = a + b$ using only limit laws.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $\lim f = 2$ and $\lim g = -5$, then $\lim(3f - g^2)$ equals

- (A) -19 (B) -7
(C) 19 (D) 31

2. Which condition is required to use the quotient law for $\lim f(x)/g(x)$?

- (A) $f(c)$ exists (B) $g(c) \neq 0$
(C) $\lim g(x) \neq 0$ (D) f and g are differentiable

3. A calculator gives $u(x) \rightarrow 1.73205$ and $v(x) \rightarrow -0.50000$. Estimate $\lim[u(x)^2 + 4v(x)]$.

- (A) -1 (B) 0
(C) 1 (D) 5

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Assume $f(x) \rightarrow L$ and $f(x)^2 \rightarrow 9$. What values of L are possible? What extra condition forces $L = 3$?

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